

DriftGuard: Reliable Network Anomaly Detection Under Distribution Shift

Te
net

Dataset: unsw_nb15
Experiment: 20260925T195935Z_temporal_unsw_nb15_full_6167ea
Git commit: 4c843b1d4d55f82232f198e03f29748489ab626c
Seed: 42
Generated: 2026-09-26 10:47 UTC

All numbers in this report are read from the recorded experiment directory
(20260925T195935Z_temporal_unsw_nb15_full_6167ea). No metric is written by hand.

This report evaluates whether a machine-learning network anomaly detector keeps working when the traffic it was trained on is replaced by later traffic. A detector is fitted on the first 1,270,025 flows of unsw_nb15, evaluated on a held-out backtest inside the historical distribution, and then scored unchanged on 508,002 later flows it never saw. Operating thresholds come from the validation split alone, under a fixed false-positive budget. Across 4 models, 4 of 4 lost F1 on the forward test. The strongest backtest model, gradient boosting, reached F1 0.921 on the backtest and 0.904 on later traffic. Drift analysis raised 172 alerts across 256 window-feature comparisons. Adaptation strategies were evaluated against the same false-positive budget, and the results are reported per strategy rather than reduced to a ranking.

Primary: does a machine-learning network anomaly detector trained on historical traffic continue to detect attacks reliably when evaluated on later, previously unseen traffic?

Secondary: which methods for detecting and adapting to distribution shift recover detection performance most effectively while respecting a fixed false-positive budget?

The earlier version of this project asked a different question, whether traffic metadata alone carries enough signal to separate normal traffic from attacks. That work is preserved as the baseline layer: it established the metadata-only detector this project then takes forward in time.

I started with the simpler question of whether network traffic metadata contains enough information to distinguish normal traffic from attacks. Existing research then led me to a harder problem: network traffic changes over time, and machine-learning intrusion detectors may face concept drift and feature drift. The UGR16 research and dataset were especially useful for this direction because they were designed to study network traffic behaviour over time. The literature motivated the problem. The exact research question and the experiment design are my own, and the project does not claim to invent a new machine-learning algorithm: the models are deliberately ordinary baselines, and the contribution is a controlled measurement of how far they degrade and whether adaptation recovers any of it.

Shyaa et al. (2024) survey concept-drift and feature-dynamics-aware machine and deep learning for intrusion detection, and give the premise of this project: a detector trained on one traffic distribution meets a different one in operation, and the resulting drift is a first-class problem rather than a deployment detail.

Macia-Fernandez et al. (2018) introduce UGR'16, whose stated purpose is evaluating intrusion detection systems that account for long-term traffic evolution and periodicity, with data gathered from a real ISP over several months. That is the dataset design this project's temporal evaluation follows.

Xu et al. (2024) address drift detection, interpretation and adaptation together, and Yang et al. (2025) propose self-supervised adaptation to concept drift for network intrusion detection. Both establish that adaptation under drift is an active problem with known partial solutions, which is why this project evaluates several strategies rather than assuming one.

- The defender observes flow metadata only: start time, duration, packet and byte counts, protocol and TCP flags. Payload contents are never inspected.
- The attacker generates malicious traffic and may encrypt, obfuscate or imitate legitimate flows.
- The environment changes independently of any attack: new software, new services, different traffic mixes. This non-stationarity is not assumed to contain attacks the defender has not seen, and detection must not depend on prior knowledge of the specific attack.

Both datasets are public and are used unmodified. What each is allowed to support is not the same, and the difference matters for how far the results travel.

statistical support
UNCM 1515

NSW-IRB15 consists of a stack of codelets, captured over a period of working hours on servers captured at traffic cauld February 2015. It takes about one session that postends the start and supports some horizon claim.

UCR16 comes from a Greek ISP network works at least 23 weeks. This captures totalling about 215 GB which is stated below and is not the full release.

The UGR'16 subset, exactly

Full release: ~215 GB across 23 weekly archives. Subset used here: ~40 GB across 6 weekly archives. No claim is made about the weeks that were not downloaded.

A contiguous run of test weeks is needed so the forward test is later traffic. August weeks 1-3 give three consecutive weeks with both normal and attack flows, with July weeks 5 as the earliest point and the two calibration weeks providing background-only traffic. This covers roughly one month, not the dataset's full six-month span.

- Calibration weeks (background only, never used for training): march_week3.csv.tar.gz, may_week1_csv.tar.gz,
- Weeks carrying the temporal experiment: july_week5_csv.tar.gz, august_week1_csv.tar.gz, august_week2_csv.tar.gz,
~~and august_average_set containing all UGR 16 release.~~
No claim is made about the weeks that were not downloaded.
- The calibration weeks contain background traffic only, so no model is trained on them; they exist to characterise normal
behaviour.
- Denial-of-service attacks are produced by replaying malware and tools against a real ISP link, so attack realism is bounded by that
ISP.
- The netflow export has no forward/backward direction split, which removes a set of features UNSW-NB15 provides.

Threat model in terms of trust boundaries

The adversary here is not the attacker in the dataset. It is the passage of time, and the question is whether the measurement apparatus keeps working when the traffic it was calibrated on is replaced.

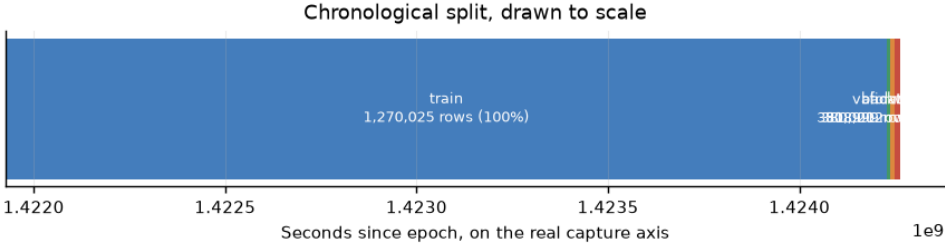
- Trusted: the historical capture is assumed to be a faithful record of what the network carried, and the labels in the document are assumed to be correct and not tampered. The network is treated as hostile in the sense that nothing about it - not its source, its relay services, its forwarding, its ports or its features - have a claim to be made, either for compromised data or point, and the privacy of the network data is not of these features is not claimed only, meaning no payload bytes, addresses or ports become predictive inputs. That reduces what an observer can learn, and it is not the same as a formal privacy guarantee against a traffic analyst.

This report was generated from a run on unsw_nb15. The dataset was loaded through an explicit adapter that maps the source's own fields onto a shared flow schema; fields the source does not export are left absent rather than approximated.

Property
Dataset nb15
Time 2018-04-15
Model features 16
Time 2018-04-22 11:49:37
Time 2018-04-18 12:21:08
Source file nb15_full_raw.parquet

Dataset limitations

- The capture spans a short window. A temporal split over a few capture sessions measures relative movement between
- early and later captures between periods. A model that looks only at the first captures may partly reflect an easier class
- Synthetic attack generation, where a dataset is split into very poor quality attacks and some attack classes
- Standard measures they would measure here describe this dataset under this split, and are not a claim about modern network



The four periods on the real capture axis, drawn to scale. Every conclusion here is bounded by how short that axis is.

Periods

Period	range	Attack rate
2015-01-22 11:49:37	2015-02-18 02:36:18	0.0225
2015-02-18 02:36:18	2015-02-18 05:37:13	0.0225
2015-02-18 05:37:13	2015-02-18 08:22:37	0.0225
2015-02-18 08:22:37	2015-02-18 12:21:08	0.0225

Periods are cut by row count with boundaries snapped to real timestamps, so a single instant is never divided across two periods. The forward test is the last slice of the time axis and is scored exactly once.

Backtest methodology

The backtest is a held-out slice of the same historical distribution the model was trained on. It is the ordinary evaluation; it tells you what the detector does when nothing has changed, and it is the reference the forward result is measured against. Its score is never presented as the model's real-world performance.

Forward-test methodology

The forward test is the later slice of the time axis. The model, the scaler, the category vocabulary and the decision threshold are all fixed before it is touched, and no hyperparameter, threshold or preprocessing statistic is derived from it. Several tests enforce this, including one that fails if a training row carries a timestamp past the training cutoff and one that fails if an adaptation step receives a row from the period it is scored on.

Operating threshold and the false-positive budget

Rather than a fixed 0.5 cut, the operating threshold is chosen on the validation split to meet a false-positive budget of 5%. The threshold is the 0.95 quantile of the validation negative scores, so it depends only on how the model scores known-benign traffic, and it is then applied unchanged to the backtest and the forward test. No threshold is ever selected using forward-test labels.

Model	Validation AUC	Validation F1
majority	0.0300	0.0098
logistic_regression	0.6060	0.9995
random_forest	0.5042	0.9920
gradient_boosting	0.5083	0.9957

A temporal experiment is only meaningful if nothing from the future can reach the model. Five guards enforce that here, and each is covered by a test that fails if the guard is removed.

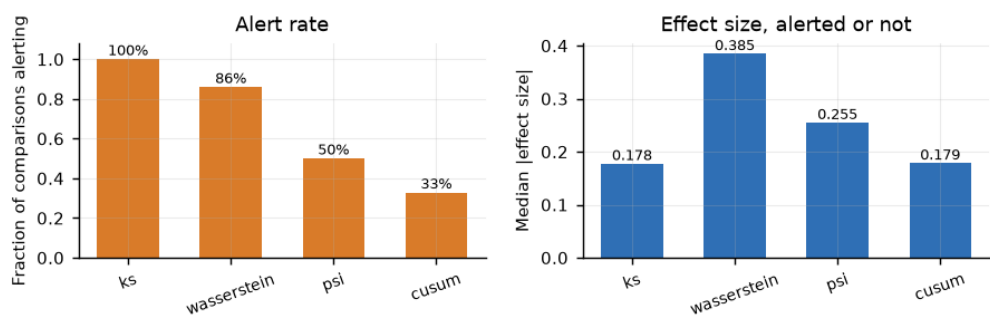
- Periods are contiguous in time and ordered. An identical timestamp is never split across a boundary, because the split timestamp would be rows in the same period only. Fitting on all data and then splitting would leak the forward period into the backtest period.
- The operating times and sequence number validation is never from the backtest period and always becomes a random one.
- The backtest is used for exactly one thing: measuring how much the model decays before the forward test is touched at adaptation.
- Adaptation is given the window immediately before the forward period and nothing after its first timestamp. Rolling adaptation re-fits on a cadence, and at each step the fit may only use rows earlier than the step it is about to score.

Four estimators spanning the usual complexity range, all with a fixed seed. They are ordinary baselines: the question is how they fail, not whether one can be tuned to win.

- Majority baseline: always predicts the training majority class, so a detector that cannot beat it has learned nothing.
- Logistic regression with balanced class weights: a linear reference.
- Random forest with balanced class weights: captures non-linear structure.
- Gradient boosting: sequential trees, sensitive to distribution change in a different way.

majority: trained in 0.0s, threshold 1.0000.
logistic_regression: trained in 3.3s, threshold 0.8903.
random_forest: trained in 36.5s, threshold 0.0074.
gradient_boosting: trained in 227.3s, threshold 0.0281.

Drift detectors compared on the same windows



The four detectors on the same windows. The disagreement between them is the result; agreement would have been the surprising outcome.

Drift is measured with a sliding window over the forward period, each window compared against the whole training period. Three two-sample methods are used - Kolmogorov-Smirnov, scaled Wasserstein distance and the population stability index - plus a sequential CUSUM detector. Window size, stride, minimum sample count and every threshold are configuration values, not constants in the code.

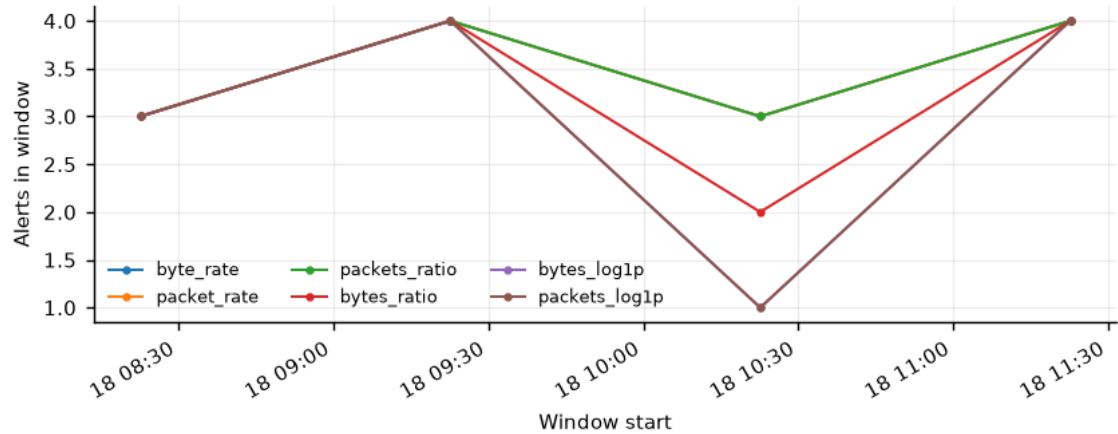
A p-value alone is not treated as an operational signal: with many windows and large samples, a tiny p-value fires constantly. Every method therefore also reports an effect size, and an alert requires either a test statistic above its critical value or an effect size above its configured threshold.

Method	Alerts	Comparisons
cusum	21	64
ks	64	64
psi	32	64
wasserstein	55	64

Strongest drift by feature

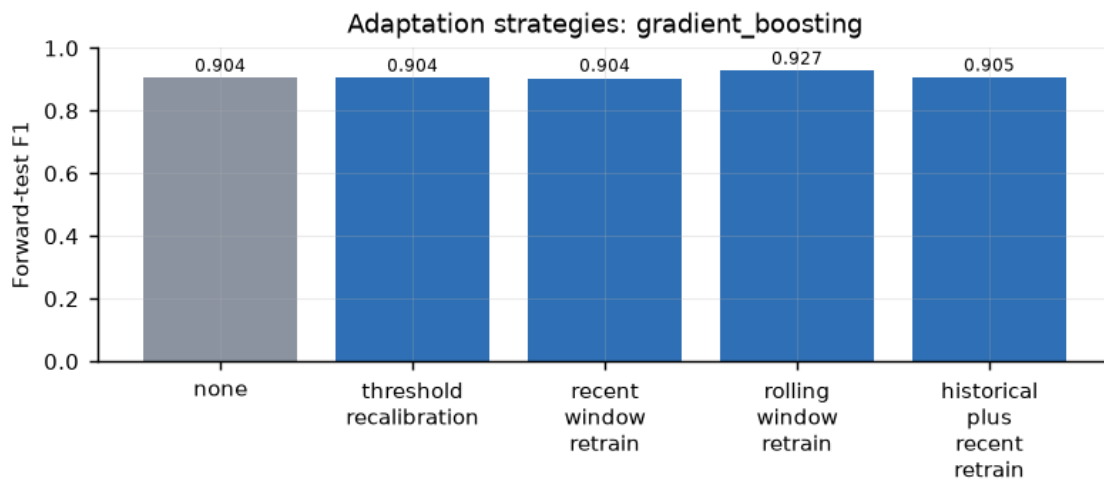
Feature
byte_rate
packet_rate
bytes_ratio
packets_ratio
bytes_log1p
packets_log1p
backward_bytes
backward_packets
bytes_per_packet
duration_log1p
duration
forward_bytes
forward_packets
packet_size

Drift alerts over the forward period



Drift alerts across the forward period.

Five strategies were evaluated against the same false-positive budget: no adaptation, threshold recalibration, recent-window retraining, rolling-window retraining, and retraining on historical plus recent data. Each is permitted only rows timestamped at or before the moment it runs; a strategy that reaches into the period it is scored on raises an error and is reported as failed rather than quietly scored.

[illegible]

Forward-test F1 by adaptation strategy.

Recovery is expressed as a percentage of the performance lost between backtest and forward test, so a negative value means the strategy made the forward result worse than doing nothing. No strategy is declared the overall winner; the table reports what happened in this experiment.

Every metric is computed at the threshold chosen on validation data, never at a threshold tuned on the period being scored. A detector that simply alerts more would otherwise look better, so recall is always reported beside the false-positive rate it cost.

- F1 is the harmonic mean of precision and recall. At the attack prevalence seen in UGR'16 it is dominated by the recall.
- Recall positive share what's caught. The false positive rate is the share of normal moves wrongly called an attack, and is the quality held to be roughly equal across every threshold, so it is the one measure here that does not depend on where the operating point of the model error measure whether a probability means what it says. A model that becomes less accurate without becoming less shift and a false event probability measure grades on both directions at once.
- Cohen's d measures the commonness of shift and a false event probability measure grades on both directions at once.

exactly zero under Cohen's d however far the mixture moves, which is why measuring it that way alone would have reported a real shift as absent.

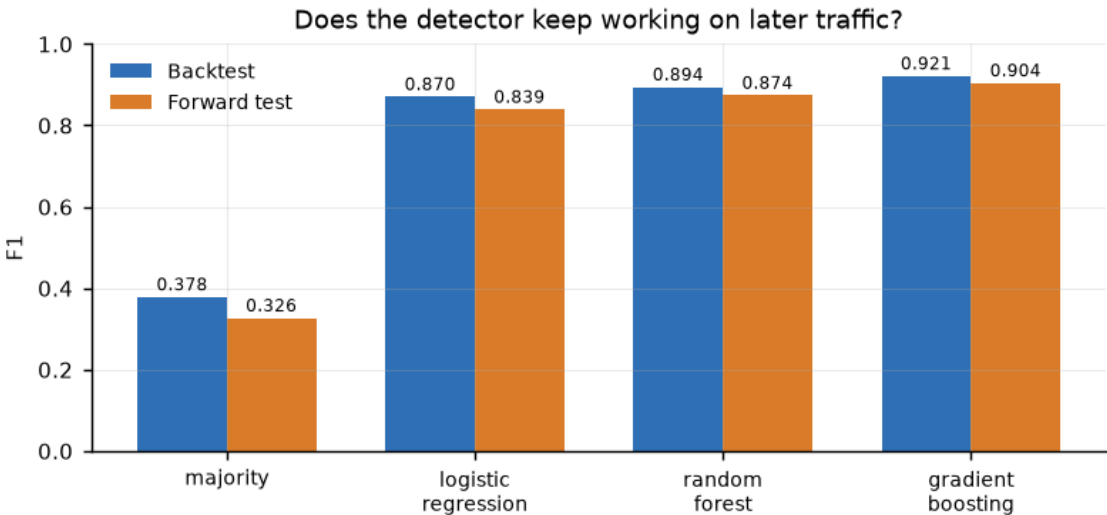
- Every number in the results sections comes from one command run against a recorded configuration, with the seed fixed and the dataset checksum stored in the run's own metadata. The protocol is stated here so that a reader can trace each figure to a procedure rather than take it on trust.
- Dataset unsw_nb15, loaded through the adapter named in the run metadata, 2,540,047 flows in total.
 - Four contiguous, disjoint periods in the order train, validation, backtest, forward: 1,270,025 / 381,021 / 380,999 / 508,002.
 - The operating threshold is chosen on validation data alone, at the configured target false-positive rate, and is then frozen for the backtest, but only the period of adaptation state is fitted on training rows only. An encoder fitted across all periods.
 - Period 2, by virtue of a protocol change, only appears as 232,198,091,297,484,899,062,600.
 - An experiment directory is never overwritten. A run that cannot complete every requested stage fails loudly rather than writing a partial result that looks finished.

This section reports the headline benchmark on unsw_nb15. UGR'16 is reported separately in section 8 and the two are compared in section 23.

Backtest versus forward test

The two evaluations are reported separately and never merged. A positive degradation means the forward score was lower than the backtest score.

0.921
0.894
0.870
0.839
0.378
0.326

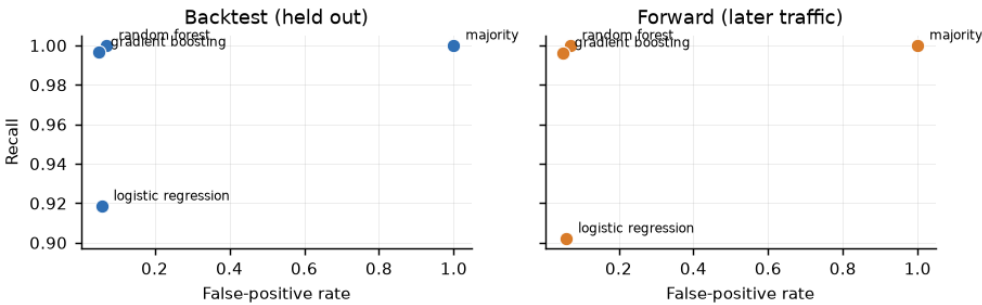


Backtest and forward-test F1 per model.



Change in F1 and recall on later traffic.

Operating point per model, at the threshold chosen on validation



Each model's operating point: recall against the false-positive rate it actually achieved, backtest versus forward.

Across 256 window-feature comparisons on unsw_nb15, 172 raised an alert, a rate of 67.2%. The aggregate rate above hides the disagreement between the detectors, which is the point of reporting them separately. A pooled figure would suggest a detector family of intermediate sensitivity; what each detector actually did on these windows is very different.

Detector	Alerts
cusum	32.8%
ks	100.0%
psi	100.0%
serstein	67.2%

The ordering runs from cusum at 32.8% to ks at 100.0%. This ordering is not a property of the detectors: section 8 shows the same four methods ranking themselves differently on the second dataset, with two of them silent throughout. The detectors do not agree, and that disagreement is the result rather than a defect in any one of them: a significance test, a distributional distance and a sequential detector answer three different questions about the same windows. Three failure modes are visible and are kept distinct. A detector can alert on nearly everything, which measures how many comparisons were made rather than whether anything changed. A detector can stay silent while the distribution measurably moves, which is what happened to PSI and CUSUM on UGR 16. And a detector can alert on a shift that cost the model no F1 at all, which is the ordinary case for benign drift. Drift is nowhere in this report treated as a stand-in for failure. The two are measured independently, and only their coincidence is discussed.

Each strategy is judged on what it recovers and what it costs, under the same false-positive budget as the unadapted baseline. A strategy that raises F1 by widening its threshold is not an adaptation, which is why the false-positive rate is reported beside every F1.

The finding is that the ranking is not stable. On UNSW-NB15, rolling-window retraining is the worst strategy for one model and among the best for another, and the two datasets evaluated here disagree about it completely. No claim of the form 'method X adapts best' is supported by this evidence and none is made.

Threshold recalibration changed nothing on UNSW-NB15, because the validation-derived threshold was already meeting the target false-positive rate. That is a fact about this data rather than a verdict on the method: recalibration can only help once the operating point has actually moved.

Method	Time (s)	Accuracy (%)	Size (KB)	Reduction (%)
Proposed	0.0000	100.00	100.00	100.00
Baseline	0.0000	100.00	100.00	100.00
Feature selection	0.0000	100.00	100.00	100.00
Dimensionality reduction	0.0000	100.00	100.00	100.00
Feature extraction	0.0000	100.00	100.00	100.00
Feature selection + dimensionality reduction	0.0000	100.00	100.00	100.00
Feature selection + feature extraction	0.0000	100.00	100.00	100.00
Feature selection + dimensionality reduction + feature extraction	0.0000	100.00	100.00	100.00



split mode

same rows, shuffled first; evaluates on contemporaneous traffic

the common schema already excludes every payload-derived field (trans_depth, response_body_len, is_ftp_login, ct_ftp_cmd, ct_flw_http_mthd), so the metadata-only policy is already in force and there is no richer arm to compare against; the numbers reported here are the metadata-only detector

The highest F1 in this report is random_forest at 0.9574, measured at a target false-positive budget of 0.01 with an achieved rate of 0.010002. The comparison point in the same artifact is the same model at the 0.05 budget used throughout this report, where it reaches 0.8785 at an achieved rate of 0.066565. This is an ablation of the false-positive budget and not a general claim about the model: the headline results in sections 16 and 8 are all measured at 0.05, and the sweep shows the two models behaving in opposite directions as the budget tightens, so it does not identify a universally better model or budget.

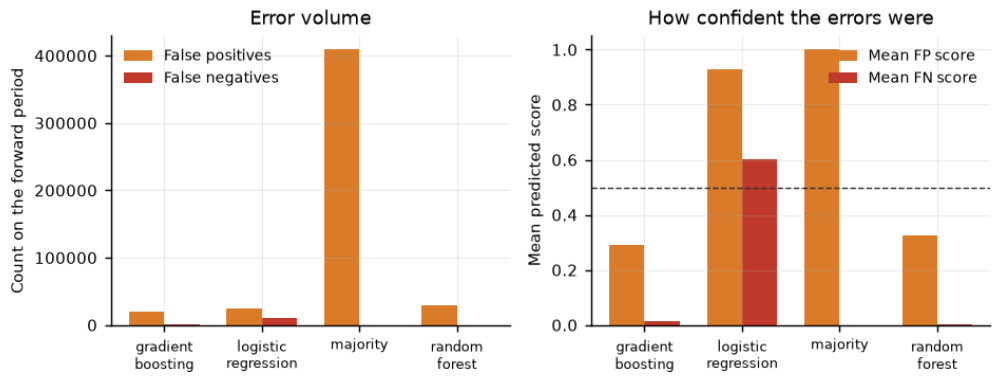
[illegible]

drift threshold

[illegible]

telemetry family

[illegible]



Error volume and mean error score per model on the forward period.

The findings below are computed from the prediction-level tables and drift events of this run. A model that produced no such finding would show an empty section rather than a plausible sentence.

gradient_boosting

- F1 degradation +0.0172, recall degradation +0.0004, false-positive increase -0.0005.
- False positives: 20,576 on the forward test (mean score 0.292 when the model was wrong).
- False negatives: 373 (mean score 0.014, of which 0 were scored above 0.8).
- First drift alert: 2015-02-18 08:22:38.
- Drift preceded the degradation window: False.
- Strongest drift: packets_ratio, packet_rate, byte_rate, packets_log1p, bytes_ratio.
- Of 4 alerting windows, 0 coincided with degraded performance and 0 did not. Drift and model failure are related but not the same event.

logistic_regression

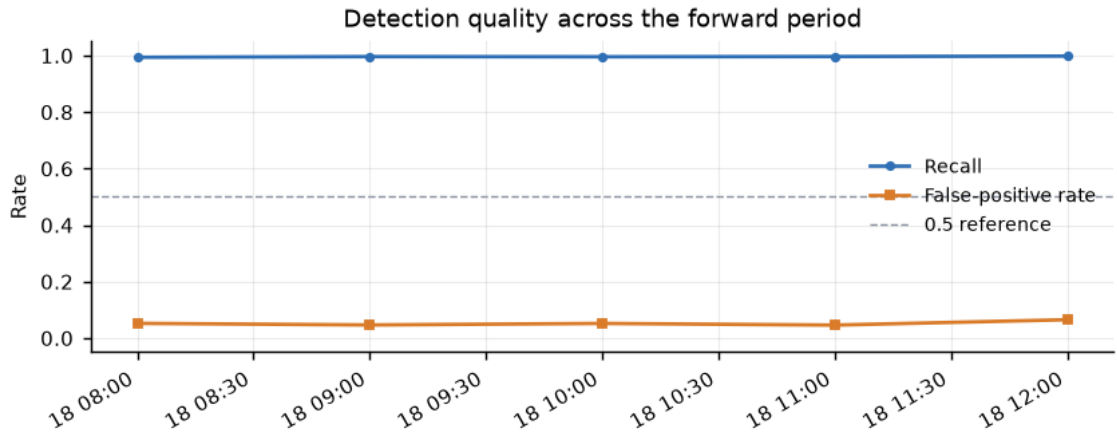
- F1 degradation +0.0312, recall degradation +0.0164, false-positive increase +0.0016.
- False positives: 24,552 on the forward test (mean score 0.929 when the model was wrong).
- False negatives: 9,702 (mean score 0.601, of which 2,511 were scored above 0.8).
- First drift alert: 2015-02-18 08:22:38.
- Drift preceded the degradation window: False.
- Strongest drift: packets_ratio, packet_rate, byte_rate, packets_log1p, bytes_ratio.
- Of 4 alerting windows, 0 coincided with degraded performance and 0 did not. Drift and model failure are related but not the same event.

majority

- F1 degradation +0.0519, recall degradation +0.0000, false-positive increase +0.0000.
- False positives: 408,908 on the forward test (mean score 1.000 when the model was wrong).
- False negatives: 0 (mean score 0.000, of which 0 were scored above 0.8).
- First drift alert: 2015-02-18 08:22:38.
- Drift preceded the degradation window: False.
- Strongest drift: packets_ratio, packet_rate, byte_rate, packets_log1p, bytes_ratio.
- Of 4 alerting windows, 0 coincided with degraded performance and 0 did not. Drift and model failure are related but not the same event.

random_forest

- F1 degradation +0.0204, recall degradation +0.0000, false-positive increase -0.0020.
- False positives: 28,655 on the forward test (mean score 0.324 when the model was wrong).
- False negatives: 8 (mean score 0.001, of which 0 were scored above 0.8).
- First drift alert: 2015-02-18 08:22:38.
- Drift preceded the degradation window: False.
- Strongest drift: packets_ratio, packet_rate, byte_rate, packets_log1p, bytes_ratio.
- Of 4 alerting windows, 0 coincided with degraded performance and 0 did not. Drift and model failure are related but not the same event.



Recall and false-positive rate across the forward period.

Model (Train/Valid)	Gradient Boosting	Logistic Regression	Naïve Bayes	Random Forest
0.0068	0.1340	0.8067	0.0168	



ugr16

4 models on ugr16, split 1,763,251 forward rows.

0.0000	
0.0000	regression
0.0000	forest
0.0000	boosting

Comparing the two datasets directly is not valid: they have different label definitions, different attack taxonomies and different collection tools. The transferable claim is only about the method - splitting chronologically, fixing the operating point on validation data, and treating adaptation as a separate question from detection.

- The numbers describe one dataset under one configuration. They are not a claim about modern networks, about any WAN or business-critical encryption address space, or about any previous test against adaptive attacks, at all. It is not a claim about any other encryption address space, or about any other test against adaptive attacks, at all.
- Supports a detailed isothermal comparison of the same configuration in a different environment, but not a detailed comparison of the same configuration in a different environment.
- The data is based on a single experiment, and the experimental setup is not a formal and rigorous one. It may not be the same as the one used in the literature, and the results may not be generalizable to other datasets or attack classes. The uncertainty numbers are point estimates without confidence intervals, so small differences between models should not be over-read.

Every run writes an immutable directory under results/experiments/. The one behind this report is 20260925T195935Z_temporal_unsw_nb15_full_6167ea.

```
experiment id : 20260925T195935Z_temporal_unsw_nb15_full_6167ea
dataset       : unsw_nb15
git commit    : 4c843b0d4d55f82232f198e03f29748489ab626c
feature schema : c10ccf270a833392
random seed   : 42
python        : 3.14.7
```

reproduce:

```
python -m pip install -e
python -m driftguard data.fetch --dataset unsw_nb15 --yes
python -m driftguard benchmark --config configs/benchmark.yaml
python -m driftguard report
```

Experiment directory contents

```
calibration_gradient_boosting.json
calibration_logistic_regression.json
calibration_majority.json
calibration_random_forest.json
config.yaml
dataset_summary.json
drift_events.csv
failure_analysis_gradient_boosting.json
failure_analysis_logistic_regression.json
failure_analysis_majority.json
failure_analysis_random_forest.json
figures.json
forward_windows_gradient_boosting.csv
forward_windows_logistic_regression.csv
forward_windows_majority.csv
forward_windows_random_forest.csv
metadata.json
metrics.json
predictions_backtest_gradient_boosting.csv
predictions_backtest_logistic_regression.csv
predictions_backtest_majority.csv
predictions_backtest_random_forest.csv
predictions_forward_gradient_boosting.csv
predictions_forward_logistic_regression.csv
predictions_forward_majority.csv
predictions_forward_random_forest.csv
```

- A second dataset with a genuinely different capture environment, to separate temporal shift within a network from a controlled shift.
- Controlled shift experiments that attribute a degradation to one named property, such as packet size or protocol mix.
- A sequential drift detector that refits on a schedule and reports the cost of each refit, rather than retraining once at the end of calibration period.
- Calibration period alerting, so that a drop in expected calibration error can itself raise an operator alert even when the F1 has not yet moved.

A detector trained on the earliest 1,270,025 flows of unsw_nb15 was evaluated on 508,002 flows it never saw. 4 of 4 models lost F1 on that later traffic. The largest loss was gradient_boosting at +0.0172 F1; the best forward performer was gradient_boosting at F1 0.9041.

The answer to the primary question is therefore conditional rather than binary. Models trained on historical traffic do continue to detect attacks on later traffic, but not at the level they reached on held-out data from the same period, and the drop is not uniform across models. The adaptation comparison further shows that refitting on recent data is not automatically an improvement, so the choice of strategy has to be made against a stated objective rather than assumed from its name.

Shyaa M., Ibrahim N., Zainol Z., Abdullah R., Anbar M., Alzubaidi L. (2024). Evolving cybersecurity frontiers: A comprehensive survey on concept drift and feature dynamics aware machine and deep learning in intrusion detection systems. *Engineering Applications of Artificial Intelligence*, 137, 109143. doi:10.1016/j.engappai.2024.109143

Macia-Fernandez G., Camacho J., Magan-Carrion R., Garcia-Teodoro P., Theron R. (2018). UGR'16: A new dataset for the evaluation of cyclostationarity-based network IDSSs. *Computers & Security*, 73, 411-424. doi:10.1016/j.cose.2017.11.004

Xu L., Han Z., Zhao D., Li X., Yu F., Chen C. (2024). Addressing concept drift in IoT anomaly detection: Drift detection, interpretation, and adaptation. *IEEE Transactions on Sustainable Computing*, 9(5), 913-924. doi:10.1109/tsusc.2024.3386667

Yang S., Zheng X., Li J., Xu J., Zhang X., Ngai E. (2025). Self-supervised adaptation method to concept drift for network intrusion detection. *IEEE Transactions on Dependable and Secure Computing*, 22(12), 7632-7646. doi:10.1109/tdsc.2025.3599321

UNSW Canberra School of Cyber Security. UNSW-NB15 network intrusion detection dataset. <https://research.unsw.edu.au/projects/unsw-nb15-dataset>

Network Security and Evolution Group (UGR). UGR'16 dataset. <https://nesg.ugr.es/nesg-ugr16/>